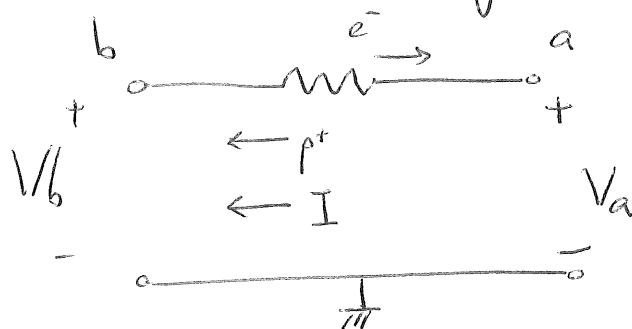


* Hoofdstuk 1

Lesing 1.2



* Stroom: stroom van positiewe lading:

$$\rightarrow i = \frac{dq}{dt}$$

$q \rightarrow [C] \rightarrow$ lading (Coulomb)
 $i \rightarrow [A] \rightarrow$ stroom (Ampère)
 $t \rightarrow [s] \rightarrow$ tyd (sekondes)

$$\rightarrow Q = \int_{t_0}^{t_1} i dt$$

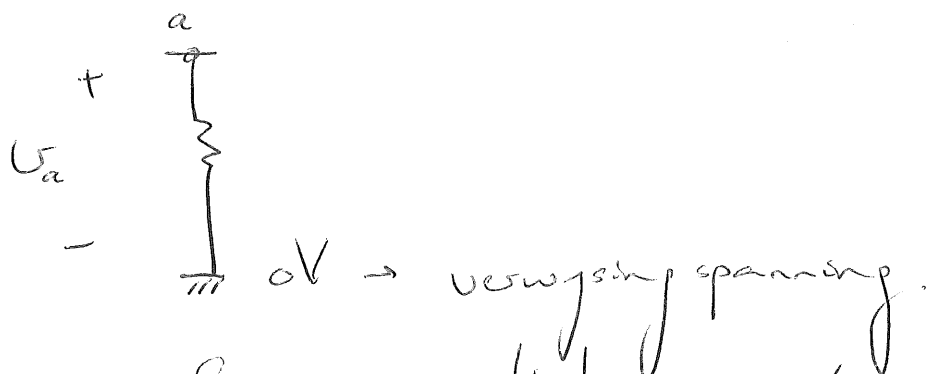
* Spanning: werk gedoen om lading te skuif.

$$\rightarrow v = \frac{dw}{dq}$$

$w \rightarrow [J] \rightarrow$ energy (Joule)

$$v \rightarrow [V] \rightarrow$$
 spanning (Volts)

$\rightarrow$ potensiaalverskil
 $\rightarrow$ spanningstal.



$\rightarrow$ Spanning altyd ~~een~~ tussen twee punte.

⇒ Drywing en Energie

(2)

* "Voltage": verandering in energie per lading.

* Wat gebeur as energie oor tyd verander?

→ Drywing: $[W] \rightarrow$ (Watts)

$$P = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt}$$

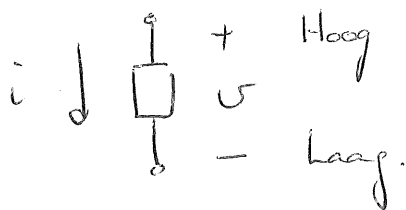
$$= V \times i$$

Drywing = Stroom $\times$ Spanning.

* DC: $P(t) = V \cdot I$ (konstant)

* AC: $P(t) = v(t) \cdot i(t)$ (Amplitude van tyd).

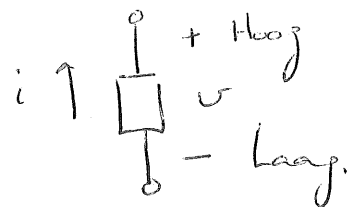
⇒ Passiewe teken konversie oor drywing.



$$P = +Vi$$

Stroom vloei van plus
na minus

~~passive~~

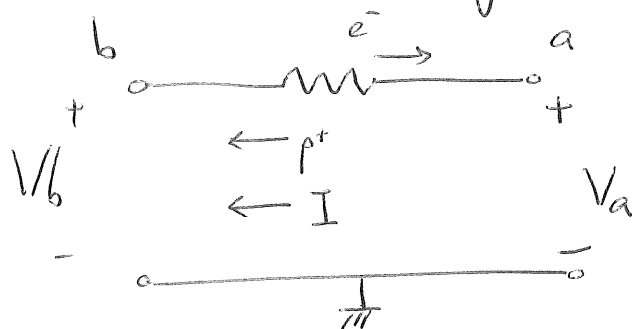


$$P = -Vi$$

Stroom vloei van minus
na plus.

* Hoofdstuk 1

Lesing 1.2



* Stroom: stroom van positiewe lading:

$$\rightarrow i = \frac{dq}{dt}$$

$q \rightarrow [C] \rightarrow$ lading (Coulomb)

$i \rightarrow [A] \rightarrow$ stroom (Ampère)

$t \rightarrow [s] \rightarrow$ tyd (sekondes)

$$\rightarrow Q = \int_{t_0}^{t_1} i dt$$

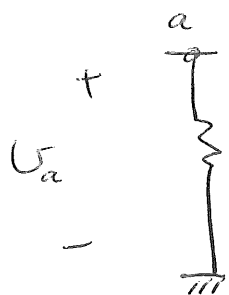
* Spanning: werk gedoen om lading te stuur.

$$\rightarrow v = \frac{dw}{dq}$$

$w \rightarrow [J] \rightarrow$ energy (Joule)

$v \rightarrow [V] \rightarrow$ spanning (Volts)

$\rightarrow$ potensiaalverskil
 $\rightarrow$ spanningsval.



OV $\rightarrow$ verwysing spanning.

$\rightarrow$ Spanning altyd ~~een~~ tussen twee punte.

⇒ Drywing en Energie

(2)

* "Voltage": verandering in energie per lading.

* Wat gebeur as energie oor tyd verander?

→ Drywing: $[W] \rightarrow$ (Watts)

$$P = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt}$$

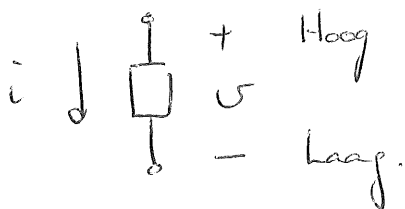
$$= V \times i$$

Drywing = Stroom $\times$ Spanning.

* DC: $P(t) = V \cdot I$ (konstant)

* AC: $P(t) = v(t) \cdot i(t)$ (funksie van tyd).

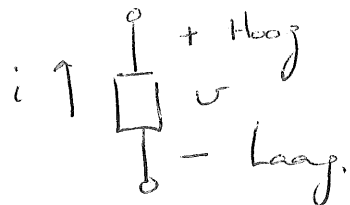
⇒ Passiewe teken konvensie oor drywing.



$$P = +Vi$$

Stroom vloei van plus
na minus

~~proom~~



$$P = -Vi$$

Stroom vloei van minus
na plus.

* Indien

$p > 0$: krag geabsorbeer

(3)

$p < 0$: krag gelewer.

Aanname: Drywing word gestryf asof
die element krag absorbeer:

→ $P = 10 \text{ W}$ → 10 Watt of 10 J/s
word deur element gebruik.

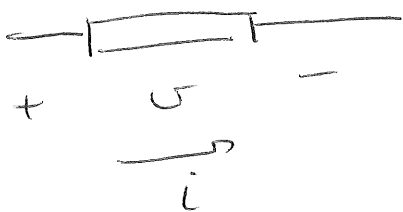
→ $P = -10 \text{ W}$ → -10 Watt of -10 J/s
word deur element gebruik,
of 10 Watt of 10 J/s
word deur element verskat.



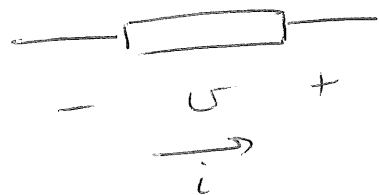
Teikenkonvensie vir drywing:

① spanning tekens

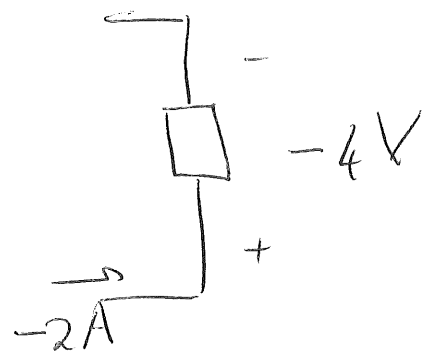
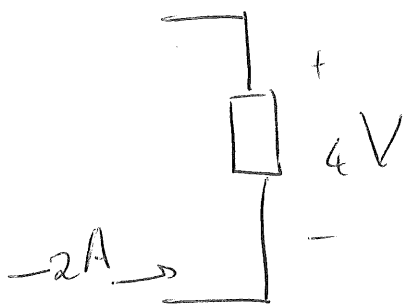
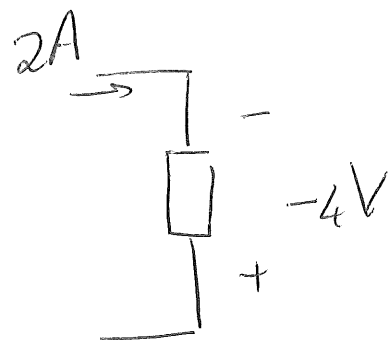
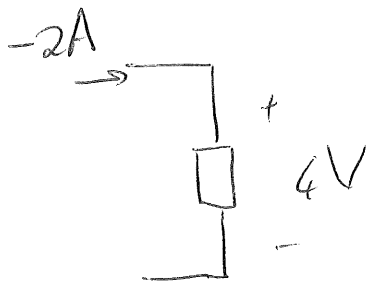
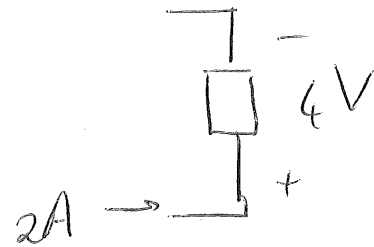
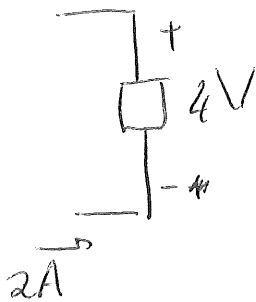
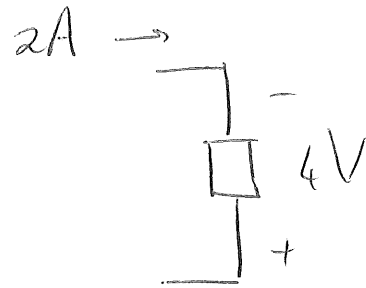
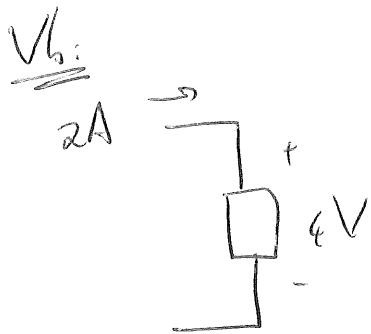
② stroom rigting.



$$p = +vi$$



$$p = -vi$$



→ Behoud van energie
∴ Behoud van drywing.

$$\sum p = 0.$$

$$(\therefore P_{\text{gelewer}} = P_{\text{gebruik}})$$

→ Energie: kapasiteit om werk te doen.

$$* p = \frac{dw}{dt}$$

$$W = \int p \, dt \quad (J = W \times s)$$

* Energie: $[J] \rightarrow \text{Joules}$

$$1 \, \text{Ws} = 1 \, \text{J}$$

$$1 \, \text{Wh} = 3600 \, \text{J} = 3.6 \, \text{kJ}$$

→ Kringbaan elemente:

* Passiewe elemente.

→ Geen energie van sy oie.

R

C

L

* Aktiewe elemente.

→ Voorsien energie!

→ Bronne!